rb_vi_bench

Reduced-basis dual cones for variational inequalities:
algorithms, metrics, and measured results

Benchmark report

September 1, 2026

Abstract

This report documents the benchmark assembled from three source repositories for reduced-basis (RB) model order reduction of contact problems. It states the problem setting, every selection algorithm under test, the exact computational definition of every metric, and the results measured on the three retained contact datasets. The object under study throughout is the *dual* cone: the reduced set from which Lagrange multipliers are drawn, on which non-negativity must hold. Sources: [BEE20] = Benaceur, Ern, Ehrlacher (`hal-02081485v2`); [NDEE22] = Niakh, Drouet, Ehrlacher, Ern (ESAIM M2AN, 2022). Both papers number their algorithms and equations independently, so every citation below carries its paper tag.

1 Problem setting

The high-fidelity (HF) problem is a discrete variational inequality of contact type: find the displacement $u(\mu) \in \mathbb{R}^N$ and multiplier $\lambda(\mu) \in \mathbb{R}^m$ solving the saddle-point system

$$A\,u(\mu) + B(\mu)^\top \lambda(\mu) = f(\mu), \qquad B(\mu)\,u(\mu) \le g(\mu), \qquad \lambda(\mu) \ge 0, \tag{1}$$

with complementarity between the last two. The sign condition $\lambda \ge 0$ is physical: contact pressure pushes, it does not pull.

Reduction replaces $\mathbb{R}^m$ by a low-dimensional set for λ. That set cannot be a *subspace*: a subspace closed under negation cannot enforce $\lambda \ge 0$. It must be a finitely generated **convex cone**

$$W_R^+ \;=\; \mathrm{span}_+\{\xi_1,\dots,\xi_R\} \;=\; \Big\{\textstyle\sum_{r=1}^R c_r\,\xi_r \;:\; c_r \ge 0\Big\}, \qquad \xi_r \ge 0. \tag{2}$$

With $\xi_r \ge 0$ and $c_r \ge 0$, membership of the cone *implies* the sign condition. This is the single structural reason the whole family of methods exists, and it is why POD is inadmissible here: its modes have mixed signs ([BEE20] §5).

Two derived objects are used repeatedly. The **full snapshot cone** $K_{\text{full}} = \mathrm{span}_+\{\theta_1,\dots,\theta_Q\}$ is everything the training multipliers generate; and the **non-negative orthant** $W^+ = \{x \in \mathbb{R}^m : x \ge 0\}$ is the largest admissible set of all. Every method produces some W_R^+ with $W_R^+ \subseteq W^+$, but—importantly—*not* necessarily $W_R^+ \subseteq K_{\text{full}}$.

The **cone projection** $\Pi_K(x) = \arg\min_{y\in K} \|x-y\|$ is computed by non-negative least squares (NNLS) on the generator matrix. It is non-linear in x, which is what makes every error below more expensive than a linear projection and is the unit in which offline cost is counted.

2 Algorithms under test

All greedy methods are *hierarchical*: the cone at cardinality R is a sub-cone of the cone at $R+1$. This is verified rather than assumed (§4). NMF is not hierarchical, and that difference is visible in several metrics.

2.1 CPG — Cone-Projected Greedy ([BEE20], Algorithm 2)

Start from $K_0 = \{0\}$. At step r, select the training snapshot worst represented by the cone built so far, and append it unchanged:

$$q_r \in \arg\max_q \left\|\theta_q - \Pi_{K_{r-1}}(\theta_q)\right\|, \qquad K_r = \mathrm{span}_+\{\theta_{q_1}, \dots, \theta_{q_r}\}. \tag{3}$$

Because $K_0 = \{0\}$ gives $\Pi_{K_0} \equiv 0$, the first step reduces to $q_1 \in \arg\max_q \|\theta_q\|$, which is [BEE20] Eq. (56). The generators are *selected snapshots*, so $K_R \subseteq K_{\mathrm{full}}$ holds by construction.

2.2 mCPG — modified CPG ([NDEE22], Algorithm 2)

Same selection loop, but the appended generator is a *residual* rather than the snapshot: $\nu_r = (\theta_q - \Upsilon_r)/\|\cdot\|$ with $\Upsilon_r \in K_{r-1} \cap (\theta_q - W^+)$. The second constraint forces $\nu_r \geq 0$, i.e. membership of W^+, which is the property the method needs. It does *not* force membership of K_{full}: a difference of two elements of $\mathrm{span}_+\{\theta\}$ need not lie in $\mathrm{span}_+\{\theta\}$. This is measured directly (§5).

2.3 ADG — Batch Normalized Angular-Defect Greedy

Not in either paper; contributed by the `greedy_algos` source. Two departures from CPG:

- **Angular selection.** It ranks candidates by the angle between a snapshot and its cone projection, $\theta_K(x) = \angle\big(x, \Pi_K(x)\big)$, rather than by residual magnitude. On the normalized set $S_{\mathrm{norm}} = \{x/\|x\|\}$ the two induce the same ordering, since the residual is $\sin\theta_K(x)$; un-normalized they differ. ADG is therefore *defined* on S_{norm}, and the tolerance becomes a genuine per-snapshot relative bound $\varepsilon \in (0,1)$ rather than one shared absolute threshold $\varepsilon \max_q \|\theta_q\|$.
- **Batch admission.** Every candidate attaining the maximal defect $\theta_{\max}$ enters in one round, not just one. This is what the method's Theorem 3.5 certificate rests on.

Three ADG variants are benchmarked, each differing from the base in exactly one respect, so that any measured gap is attributable:

key	varies	against
`adg`	— (reference form)	
`adg_momentum`	stopping rule: $\lvert e(p) - e(p-1)\rvert / e(p-1) \leq \varepsilon$	`adg`
`adg_k0`	initialization: $K_0 = \{0\}$, i.e. CPG's first step	`adg`

The base ADG initializes not from $K_0 = \{0\}$ but from the *pair* of snapshots at the largest mutual angle. This is its one departure from both papers, and `adg_k0` is its ablation: the first generator is instead $\arg\max_q \|\theta_q\|$ taken on the *unnormalized* snapshots ([BEE20] Eq. (56)), normalized, after which the identical angular loop runs. Note that $K_0 = \{0\}$ does not by itself define ADG's first step: with the zero cone every defect is a right angle, all candidates tie, and batch admission would take the entire training set at once — Eq. (56) is the tie-break the papers supply. A consequence of the pair start is that base ADG has *no* $R = 1$ state; its trajectory is $R = 2, 3, 4, \dots$

2.4 Baselines

- **NMF** ([BEE20] §6.4): optimized non-negative atoms, free to sit anywhere in W^+. A genuine competitor, but cardinality-only — [BEE20] §5's objection is precisely that "the user does not specify an error tolerance but only the cardinality" — and non-deterministic in its initialization.

- **Orthant** (reference, not a competitor): canonical directions e_i selected along mCPG's own iteration, so the only difference from mCPG is the generator itself. Its pairwise angles are exactly $90°$, the widest any cone in W^+ can be. Its reported cost is the mCPG ordering run, not the (free) cone assembly.
- **POD** (*negative* control): the best unconstrained reconstruction at a given cardinality, and a direct measurement of the sign violation that motivates the cone construction. Its non-negativity check is *expected* to fail.

3 Metrics: how each is computed

3.1 Precision

For a cone W_R^+ with generator matrix Ξ, each snapshot is projected by NNLS and the residual is reported relative to a normalizer. Two normalizers are reported because they answer different questions:

$$\underbrace{\frac{\|\theta_q - \Pi(\theta_q)\|}{\max_{q'} \|\theta_{q'}\|}}_{\text{shared denominator}} \qquad \text{and} \qquad \underbrace{\frac{\|\theta_q - \Pi(\theta_q)\|}{\|\theta_q\|}}_{\text{per-snapshot}}. \tag{4}$$

The shared denominator keeps the number comparable against the tolerance that produced it; the per-snapshot form is what ADG's tolerance actually bounds. They diverge whenever snapshot magnitudes spread. Train and test are reported separately.

Non-negativity is checked on the reconstruction: `nn_max_violation` $= \max(0, -\min_i \hat{\lambda}_i)$.

3.2 Performance (offline cost)

Wall-clock is machine-dependent, so the primary cost is a *count of constrained solver calls* — NNLS, `lsq_linear` and `minimize` — captured by instrumenting the solvers during a fit. One caveat governs its reading: it counts *constrained* solves only, so NMF records exactly 0, meaning "issues none", not "is free". ADG likewise records 0 at $R = 1, 2$, because it seeds from a Gram-matrix argmin and NNLS first appears at $R = 3$.

3.3 Stability

- **Gram conditioning:** $\kappa(\hat{\Xi}^\top \hat{\Xi})$ on *column-normalized* generators. Normalizing is essential: without it the number is dominated by snapshot magnitude spread, and [BEE20] (raw generators) and [NDEE22] (normalized) would be incomparable. Undefined for $R < 2$ and reported as such.
- **Orthogonality** e_{orth} ([NDEE22] Eq. 41): $\|(I - \Pi_{K_r})(\nu_{r+1})\|$, the sine of the angular defect of each accepted generator.
- **Determinism:** across-seed spread, which is the non-determinism [BEE20] §5 raises against NMF.

3.4 Cone geometry

This family scores the cone against K_{full} rather than against the finite snapshot set, by Monte-Carlo sampling of Dirichlet(1) conic combinations.

$$\text{missed (too small)}: \quad \mathbb{E}_{x \sim K_{\mathrm{full}}} \left\| x - \Pi_{W_R^+}(x) \right\|, \tag{5}$$

$$\text{excess (too large)}: \quad \mathbb{E}_{x \sim W_R^+} \left\| x - \Pi_{K_{\mathrm{full}}}(x) \right\|, \tag{6}$$

$$\text{two-sided}: \quad \tfrac{1}{2}\big(\text{missed} + \text{excess}\big). \tag{7}$$

Neither direction implies the other: a cone can cover K_{full} perfectly while extending far beyond it, or sit strictly inside while missing most of it. The two-sided figure is the *average*, not the maximum. A maximum reports only whichever direction is larger and so silently changes which quantity it plots from one method to the next; the average is the same functional for every method, keeps both faults in view, and is still a metric on cones (symmetric, zero only when both vanish, triangle inequality inherited).

Extent (how much space the cone encloses)

Cut W_R^+ by a hyperplane $\langle x, u\rangle = h$ shared by every method, with $u = (1,\dots,1)/\sqrt{m}$ — strictly positive on $W^+ \setminus \{0\}$, so the cut is transversal for every admissible cone, and no coordinate is privileged. The section is the convex hull of $p_i = h\,\xi_i/\langle \xi_i, u\rangle$, and what is reported is its **mean width**, normalized by that of K_{full}'s section:

$$\operatorname{extent}(W_R^+) \;=\; \frac{w(S_R)}{w(S_{\text{full}})}, \qquad w(S) \;=\; \mathbb{E}_{g\sim\mathcal{N}(0,I_m)}\Big[\max_i \langle p_i, g\rangle - \min_i \langle p_i, g\rangle\Big]. \tag{8}$$

Three properties make this the right statistic, and each rules out an alternative that was tried and discarded:

- *Measured in the dimension of the space, not the number of generators.* The ξ_i are R rays in $\mathbb{R}^m$ with $R \ll m$; they are not a basis. Taking the section's $(R-1)$-volume builds the method's cardinality into the yardstick.
- *Monotone in R by construction.* $W_R^+ \subseteq W_{R+1}^+$, so the section can only grow; more vertices means a maximum over a superset, so the integrand rises pointwise in g. The Gaussian directions are drawn once from a fixed seed, so monotonicity holds per realization and Monte-Carlo noise cannot manufacture a decrease.
- *Positive on a thin cone,* where every ambient volume, solid angle or covered-sphere fraction is identically zero.

Values above 1 are meaningful: the cone is wider than K_{full}, which requires leaving it. Scale-invariance ($\operatorname{span}_+\{c\xi\} = \operatorname{span}_+\{\xi\}$) means generator normalization cannot affect it.

Aperture (conditioning, *not* extent)

Mean pairwise angle between normalized generators. Retained as a conditioning diagnostic only. It is not a measure of enclosed space: it saturates once generators are mutually well separated, and a generator added strictly *inside* the cone enlarges the region not at all while still moving the mean.

3.5 Solved error (`metrics.online`)

Every metric above scores a cone against snapshots; none solves anything. This one assembles and solves the reduced saddle-point problem ([BEE20] Alg. 1 line 4) at each held-out parameter,

$$\hat{A} = V^\top A V, \quad \hat{B} = \Xi^\top B(\mu) V, \quad \min_{\alpha\geq 0}\ \tfrac{1}{2}\alpha^\top \hat{B}\hat{A}^{-1}\hat{B}^\top \alpha - \alpha^\top\big(\hat{B}\hat{A}^{-1}\hat{f} - \hat{g}\big), \tag{9}$$

and reports the relative error in *both* components against the HF solution. The bound $\alpha \geq 0$ is the discrete form of span_+, and it is why the recovered $\hat{\lambda} = \Xi\alpha$ is non-negative exactly rather than approximately. Projection error and solved error are different functionals of the same cone: the solve minimizes an energy over W_R^+, it does not project the true multiplier onto it.

3.6 Cross-implementation agreement

Two independent transcriptions of CPG and of mCPG are retained deliberately. Agreement between them is checked at matched tolerance, so that a disagreement is a *finding* about an under-specified step rather than a rounding difference.

4 Experimental protocol

Datasets. Three contact problems on real geometry:

key	reported name	dim $\times n$	geometry
`fem_lambda`	Half-disks of Hertz	57×50	mirrored contact line
`physics`	3D Pellet-Cladding	7676×94	76×101 surface grid
`membrane_2d`	membrane (2-D)	497×125	scattered 2-D patch

Two modes, never mixed. *Matched tolerance* gives every method the same ε and reports the R it needed; *matched cardinality* gives every method the same R. Only the second is used for comparison figures, because tolerances are not commensurable across methods (ADG's is per-snapshot relative; CPG/mCPG's is one shared absolute threshold).

One greedy trajectory. The cardinality sweep re-fits at each R. That is only legitimate if a greedy stopped at R yields exactly the first R generators of a longer run, so the property is asserted by test rather than assumed — and testing it found a real violation, since the upstream fixed-component helper answered $R = 1$ for ADG through a different selection rule.

Skips are recorded, never dropped. A cell that does not run emits a `skip_reason`, because "not measured here" and "scored badly here" must never be confusable.

Axes. All figure axes are linear, with one documented exception: the Gram condition number, which is ≥ 1 by definition (so the zero-dropping failure of a log axis cannot occur) and spans eleven decades before the basis goes singular.

5 Results

All numbers below are medians over the matched-cardinality sweep $R = 1, \ldots, 40$ on the three datasets, from the current run (817 of 840 cells; 1848 s).

5.1 Non-negativity holds, exactly, for every cone method

Across all 817 cells and every cone method, the maximum non-negativity violation of the reconstruction is exactly 0. This is structural, not numerical: $c \geq 0$ and $\xi \geq 0$. The POD negative control behaves as [BEE20] §5 predicts and fails: at $R = 8$ it violates on 12.9% of entries on Half-disks and 89.4% on Pellet-Cladding. Driven through the reduced *solve*, POD's recovered multiplier reaches -195 where every cone method returns exactly 0.

5.2 Precision and cost

dataset	method	test err	solves	extent	two-sided
3D Pellet-Cladding	CPG	**0.00261**	1880	0.717	**0.00024**
	mCPG	0.00339	762	5.74	0.0430
	ADG	0.01482	**893**	0.999	0.00232
	NMF	0.01272	0	1.47	0.0489
Half-disks of Hertz	CPG	0.0599	1240	0.999	**0.00014**
	mCPG	0.0561	305	16.4	0.105
	ADG	0.0599	**589**	0.999	0.00016
	NMF	**0.0297**	0	14.4	0.0451
membrane (2-D)	CPG	0.630	4000	0.923	0.0839
	mCPG	0.625	1849	0.947	0.0871
	ADG	0.619	**1900**	0.906	**0.0532**
	NMF	**0.591**	0	1.39	0.0891

ADG costs consistently about half of CPG. The median ratio of constrained solver calls, CPG over ADG at matched R, is 2.15×, 2.20× and 2.15× on the three datasets — remarkably stable across problems differing by two orders of magnitude in dimension. On Half-disks ADG reaches the identical test error for that halved cost.
NMF wins on raw reconstruction where it wins, reproducing [BEE20]'s own honest finding, but at a cost the offline column cannot show and with cones that reach far outside K_{full} (extent 14.4 on Half-disks).

5.3 mCPG leaves the snapshot cone — and how much depends on the data

[NDEE22] Remark 4.3's parenthetical claims ν_r lies "in fact of $\text{Span}^+(\{\theta_{q_n}\})$". It does not hold as stated. The fraction of mCPG generators lying outside K_{full} (median over the sweep):

	CPG	mCPG	ADG
3D Pellet-Cladding	0.000	0.271	0.000
Half-disks of Hertz	0.000	**0.917**	0.000
membrane (2-D)	0.000	**0.951**	0.000

What *does* hold is the W^+ membership the same sentence claims first, and that is the part the construction rests on. CPG and ADG are at exactly 0 by construction, as they must be: their generators *are* snapshots. The extent metric confirms this independently — mCPG's cone is 5.7× to 16.4× wider than everything the snapshots generate, which is only possible by leaving K_{full}.

5.4 ADG's pair initialization: mostly irrelevant, and never decisive

Comparing base ADG against `adg_k0` (CPG's first step) at matched R:

dataset	pair better	$K_0 = \{0\}$ better	identical
3D Pellet-Cladding	0	0	39 of 39
Half-disks of Hertz	0	0	30 of 30
membrane (2-D)	12	**27**	0 of 39

On two of three datasets the two initializations produce *the same cone at every cardinality*, because the largest-magnitude snapshot already lies in the largest-angle pair. Where they differ, the CPG-style start wins more often. Offline cost is unaffected: the choice moves *which* snapshots are selected, never how many solves it takes. The pair initialization is therefore not earning its departure from the papers on this evidence.

5.5 Cross-implementation agreement

CPG's two independent transcriptions agree on **36 of 36** comparisons. mCPG's do not: of 18 comparisons, 7 are exactly equivalent, 1 agrees within solver tolerance, 9 produce a *different cone of the same accuracy*, and 1 is genuinely divergent. This is a finding about [NDEE22] Algorithm 2 line 9, whose constrained shift is under-specified — the two implementations resolve it with different solvers, and the resulting cones differ.

5.6 Conditioning

Median Gram condition number (column-normalized): Pellet-Cladding CPG 9.4×10^6, mCPG 1.1×10^7, ADG 8.2×10^4; membrane CPG 104, mCPG 98, ADG 103. On Half-disks every method exceeds $1/\varepsilon_{\text{mach}} = 4.5\times10^{15}$ from $R \approx 17$ onward: the basis is numerically singular there and the reported value beyond that point is round-off, not a measurement. Figures shade that region rather than let it set the axis.

6 Caveats

1. **No dataset currently drives a reduced solve.** The two 1-D obstacle toys were the only sources shipping A, $B(\mu)$, f and g; with them removed, the inf-sup family and the solved-error metric report nothing on the three retained datasets. Both remain implemented and generic, and their tests build such a problem themselves.
2. **The orthant's cost is mCPG's ordering run,** not the cone assembly, which is free. Tracking mCPG is what buys the comparability; the price is that the cost column measures the tracking.
3. **Extent above 1 is not an error.** It means the cone is wider than K_{full}, which requires leaving it. mCPG and the orthant both do.
4. **The full-cone reference in the solved-error table is not a lower bound.** The reduced solve minimizes over the cone rather than projecting onto it, so a smaller cone can land closer.
5. **Half-disks conditioning is meaningless past** $R \approx 17$, as above.
6. **Membrane errors are large in absolute terms** (≈ 0.6) for every method: 125 snapshots on 497 dual dofs with a moving contact patch is simply not compressible to $R \leq 40$. Comparisons there remain valid; the absolute level is a property of the dataset.

Reproduction

```
python -m bench.runner --datasets fem_lambda physics membrane_2d \
       --subsample 200 --out results
python -m bench.runner --datasets fem_lambda physics membrane_2d --deltas \
       --cardinalities $(seq 1 40) --no-infsup --no-determinism \
       --subsample 200 --out results/sweep_dense
python -m bench.report  --results results --out results/report.txt
python -m bench.figures --results results/sweep_dense --out results/figures --split
```

Main grid 174/210 cells in 919 s; dense sweep 817/840 cells in 1848 s; 145 tests passing.